Day 9 – Task 1 Experiment

AI Impact Assessment & Real-World Tool Evaluation

Total Time Allocation: 2 hours

- Phase 1 (Baseline): 20 min
- Phase 2 (AI-Assisted): ~70 min
- Phase 3 (Documentation): 30 min

Phase 1: Baseline (0–20 minutes)

Establish Your Control

Perform the task normally with **zero AI assistance**. Raw human work.

While executing, keep a scratch note of:

- Where you slowed down or got stuck
- Where you hesitated or lacked confidence
- Where you felt the urge to search or look something up
- Where you made a decision without full certainty

This friction map becomes your baseline. *If you don't feel friction during this phase, you've chosen the wrong task—it should be something that challenges you.*

Baseline Metrics to Capture

Choose **one primary metric** and track it consistently:

Metric	How to Measure	Example
Time	Use a timer; record minutes	18 min for task completion
Quality Score	Rate output 1–10 based on pre-defined criteria	7/10 for clarity, accuracy, completeness
Completion Rate	% of task done to acceptable standard	85% finished within time limit
Confidence Level	1–10 self-assessed confidence in output	6/10 confidence in correctness

Record this in your log:

Baseline Time: ___ minutes
Quality/Confidence Score: ___ / 10
Primary Friction Points: (list 2–3 specific moments)

Phase 2: AI-Assisted Version (20–90 minutes)

Run the Same Task with Strategic AI Support

The Core Rule: Stop the moment you achieve ~10% improvement on your chosen metric.

Why? Because beyond this threshold, effort-to-gain ratio flips negative. Real-world AI adoption fails when people chase diminishing returns.

How to Use AI (With Guardrails)

Use AI for:

- Drafting initial versions or frameworks
- Structuring messy thoughts or raw data
- Explaining concepts or clarifying context
- Brainstorming alternatives or approaches
- Checking blind spots or reviewing logic
- Surface-level iteration on wording, syntax, or presentation

Do NOT let AI:

- Make final decisions for you
- Invent facts or details without verification
- Replace your domain judgment or expertise
- Become a black box (you must understand each step)

Treat it like: A junior intern with infinite stamina and zero accountability. Useful for heavy lifting, but you own the output.

Track During This Phase

As you work, note:

- Which AI tools you use (Claude, Perplexity, ChatGPT, etc.)
- Specific prompts or approaches that worked

- Moment you hit the 10% improvement threshold
- Any outputs you had to correct or reject

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AI Tool(s) Used: ___
Key Prompts: (e.g., "Generate a first draft of...")
Time Improvement Observed: ___ minutes (original: 18 min → now: ___ min)
Quality/Confidence Improvement: ___ → ___ / 10
Stopped at 10% gain: YES / NO (if not, why not?)
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Phase 3: Documentation (90–120 minutes)

The Real Work—Synthesize Your Findings

This is where shallow observations become usable insights. Treat it seriously.

📋 Experiment Log — Day 9

Task Chosen

(Be specific. Not "coding" but "write JWT auth middleware in Node.js for a Next.js app")

Task Description: [2–3 sentences explaining what you actually did]

Baseline Performance (No AI)

Time taken: ___ minutes

Subjective difficulty: Low / Medium / High

Quality/Confidence score: ___ / 10

Most frustrating moment: [One sentence. What blocked you most? Where did you feel lost?]

Friction map:

- [Friction point 1]
- [Friction point 2]
- [Friction point 3]

AI-Assisted Version

Tool(s) used: [ChatGPT / Claude / Perplexity / Other]

How AI was actually used: [Be concrete. Not "I used ChatGPT" but "Generated first draft with prompt X, then refined structure using Claude"]

Improvement observed:

- Time saved: ~___ minutes (original ___ → now ___)
- Quality change: Better / Same / Worse
- Confidence change: Higher / Same / Lower
- Completion rate: ___% → ___%

⛔ **Stopped at 10% improvement here?** YES / NO

If no: Why did you continue? What was the cost?

What AI Helped With

Be concrete and specific. No fluff.

Examples of good entries:

- "Reduced blank-page inertia—took me 2 min to go from 'I don't know where to start' to 'okay, here's a structure'"
- "Generated 5 alternative approaches in 90 seconds; I picked the best one"
- "Explained async/await concepts in plain language when I was confused"
- "Checked my regex logic for edge cases I'd missed"
- "Iterated on variable names and function signatures 10x faster"

Examples of weak entries (avoid):

- "Made things faster" (too vague)
- "It was helpful" (no insight)
- "Generated code" (doesn't explain what value this added)

Your list:

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What AI Couldn't Do

This is the gold. This is where you learn what these tools actually can't handle. Be honest.

Common categories (pick what applies):

- Couldn't understand your specific context or constraints
- Gave "technically correct, practically useless" answers
- Missed edge cases or real-world complications
- Sounded confident while being shallow or incorrect
- Required heavy correction or rework
- Lacked domain-specific intuition or reasoning
- Failed on long-context or multi-step reasoning
- Made assumptions that violated your requirements

Your observations:

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Why It Failed There

This is where you prove you're thinking like an engineer, not just a user.

For each limitation above, explain the root cause. Frame it as:

- **Lack of context:** [Tool didn't have access to X information it needed]
- **No real-world state:** [Tool can't know about live system behavior, user data, etc.]
- **Optimized for plausibility:** [Answer sounded right but was actually wrong]
- **Training data overgeneralization:** [Tool extrapolated from training data without understanding nuance]
- **Weak causal reasoning:** [Tool can't reason deeply about why something happens]
- **Structural limitation:** [This task type is fundamentally hard for LLMs]

Your analysis:

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Verdict (1–2 Lines)

Write a concise summary of when/how to use this tool for this task type:

"AI is useful for ___, unreliable for ___, and dangerous when ___."

Example: "Useful for initial scaffolding, unreliable for security logic, dangerous when you treat its confidence as correctness."

The Meta-Insight

You're not training AI usage. You're training AI judgment.

Anyone can prompt ChatGPT. Very few people can say:

- When to stop using AI (the 10% rule)
- When not to trust the output
- What task types structurally confuse these tools
- When using AI costs more time than it saves

This is senior-level thinking. Document it.

Your insight: [What did you learn about using this tool well?]

Measurement Reference Table

Track your metrics here for quick visual comparison:

Metric	Baseline	AI-Assisted	Improvement	Stop?
Time	___ min	___ min	___%	YES/NO
Quality	___ / 10	___ / 10	___%	YES/NO
Confidence	___ / 10	___ / 10	___%	YES/NO

Hit 10% on any metric? That's your stopping point.

Post-Experiment Checklist

- [ ] Completed baseline (20 min, felt friction)
- [ ] Ran AI-assisted version (stopped at ~10% improvement)
- [ ] Documented "What AI Helped" (concrete, no fluff)
- [ ] Documented "What It Couldn't Do" (honest gaps)
- [ ] Explained "Why" (root cause analysis, not surface-level)
- [ ] Written the Verdict (1–2 sentences)
- [ ] Captured the meta-insight (what did you learn about tool judgment?)
- [ ] Recorded metrics in table for comparison

Notes for Your Evaluation Project

This experiment builds a case study for your LLM evaluation framework. Use these findings to:

- Categorize task types where AI adds real value vs. hype
- Identify blind spots in AI reasoning for your domain
- Document when AI confidence is most dangerous
- Build a personal "AI judgment" rubric for future tool selection

The 10% threshold filters out productivity theater. Real gains are measurable.